Lecture 18

Resolvents as analytic functions

Objectives. *A bounded operator has a holomorphic inverse away from its spectrum. Contour integrals of that inverse give projections and analytic simple eigenvalue branches. For a bounded Friedrichs model we derive the scalar resolvent, continue it across an interval and prove its leading resonance decay rate.*

By the end of this lecture you will be able to:

- Prove the resolvent identity and its norm estimates.
- Construct contour projections and a simple eigenvalue branch.
- Recover a supplied density from its Stieltjes transform.
- Derive the continued scalar pole of the bounded Friedrichs model.

18.1 Bounded operators and their resolvents

Let $\mathcal{H}$ be a complex Hilbert space. Its inner product is antilinear in the first slot. A linear map $A : \mathcal{H} \to \mathcal{H}$ is *bounded* if its operator norm $\|A\| = \sup_{\|u\|\le 1} \|Au\|$ is finite. It is *self-adjoint* if $\langle Au, v\rangle = \langle u, Av\rangle$ for all u, v. All operators in this lecture are bounded and defined on the whole space.

The space $\mathcal{B}(\mathcal{H})$ of bounded operators is complete in operator norm. Indeed, for a norm-Cauchy sequence A_n, each $A_n u$ is Cauchy in $\mathcal{H}$. Its limit defines a linear map Au. The estimate $\|(A_n - A_m)u\| \le \varepsilon \|u\|$ passes to the limit in m, proving boundedness and norm convergence to A. Also $\|AB\| \le \|A\| \, \|B\|$. Hence, if $\|B\| < 1$, the geometric series converges in operator norm and multiplication of its partial sums gives

$$(I - B)^{-1} = \sum_{n=0}^{\infty} B^n.$$

Both products with $I - B$ equal I, since the remainder B^{N+1} tends to zero.

The *spectrum* $\sigma(A)$ consists of the $z \in \mathbb{C}$ for which $A - zI$ has no bounded two-sided inverse. Outside it define the *resolvent* by $R_A(z) = (A - zI)^{-1}$. The density letter ρ is not used for a set of spectral parameters. Here norm holomorphy means a locally convergent operator-norm power series.

Theorem 18.1. *The complement of $\sigma(A)$ is open, and the resolvent is holomorphic in operator norm there. For two points where it exists,*

$$R_A(z) - R_A(v) = (z - v) R_A(z) R_A(v), \qquad R_A'(z) = R_A(z)^2.$$

For self-adjoint A, every nonreal z has a resolvent and $\|R_A(z)\| \le 1/|\operatorname{Im} z|$.

Proof. Multiplying the difference of the two inverse equations gives the resolvent identity. The factors commute because $A - zI$ and $A - vI$ do. Near a point v with inverse, factor $A - zI$ to obtain

$$R_A(z) = \sum_{n=0}^{\infty} (z - v)^n R_A(v)^{n+1}, \qquad |z - v| \, \|R_A(v)\| < 1.$$

This proves openness, norm holomorphy and the derivative formula. For example, if $|z| > \|A\|$, apply the geometric series to A/z to obtain an inverse of $A - zI$.

If A is self-adjoint, $\langle u, Au \rangle$ is real. Cauchy–Schwarz gives

$$|\operatorname{Im} z| \, \|u\|^2 = |\operatorname{Im} \langle u, (A - zI)u \rangle| \le \|u\| \, \|(A - zI)u\| .$$

Whenever the inverse exists this proves its stated bound. To prove existence for $z = x + \mathrm{i}y$ with $y > 0$, start at $x + \mathrm{i}Y$ with $Y > \|A\|$. The local inverse series continues down the vertical line by any step shorter than the current height, because the inverse norm is at most the reciprocal height. Repeated halving reaches a height between y and $2y$ in finitely many steps. One final step reaches y. All intermediate inverses obey the bound just proved. The same argument in the lower half-plane proves the result for $y < 0$. ∎

18.2 Contour projections and a simple branch

The notation $\operatorname{ran} P$ denotes the range of an operator. Integrals of continuous operator-valued functions on a circle are limits of Riemann sums in the complete normed space $\mathcal{B}(\mathcal{H})$. They satisfy the norm bound by arc length times the maximum norm. For fixed $x, y \in \mathcal{H}$, continuity of the inner product passes through these sums and their limit:

$$\left\langle x, \left(\int_C X(z) \, \mathrm{d}z \right) y \right\rangle = \int_C \langle x, X(z) y \rangle \, \mathrm{d}z.$$

If X has a norm power series, each scalar function on the right is holomorphic. The scalar boundary-cancellation theorem, Lemma 10.1, therefore gives contour deformation through regions of holomorphy. Equality of all these inner products gives equality of the operator integrals: if an operator Z has $\langle x, Zy \rangle = 0$ for all x, y, choosing $x = Zy$ gives $Zy = 0$. This supplies the contour calculus used below without an additional operator-valued Cauchy theorem.

A projection here means an *idempotent* operator: applying it twice has the same effect as applying it once. An orthogonal projection also requires the selected range to be orthogonal to the removed subspace. The following example separates these meanings.

Take the two-mode operator and a vector

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \qquad C = \{ z \in \mathbb{C} : |z - 1| = 1/2 \},$$

with counter-clockwise orientation. Direct inversion gives

$$R_A(z) = \begin{pmatrix} (1 - z)^{-1} & -[(1 - z)(3 - z)]^{-1} \\ 0 & (3 - z)^{-1} \end{pmatrix}.$$

The circle encloses the first pole only. Its entrywise residues give

$$P = -\frac{1}{2\pi\mathrm{i}} \int_C R_A(z)\,\mathrm{d}z = -\operatorname{Res}_{z=1} R_A(z)$$
$$= \begin{pmatrix} 1 & -1/2 \\ 0 & 0 \end{pmatrix}, \qquad Px = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad (I-P)x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The selected vector has eigenvalue one. The removed vector has eigenvalue three. Their sum is the original vector. Direct multiplication gives $P^2 = P$, but the matrix is not equal to its adjoint. Thus this projection is not orthogonal for the usual inner product on $\mathbb{C}^2$.

The spectral circle and the vector decomposition occupy different spaces in the diagram. The sloping line in the vector plane is the removed eigenspace. Projection proceeds parallel to that line.

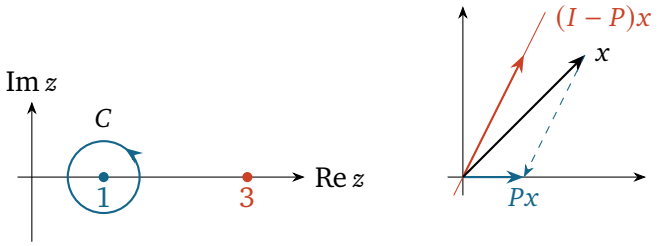


Figure 18.1: The contour selects the pole at one. In the separate vector plane, the selected and removed eigenvectors add to x; they are not perpendicular.

A parameter can move the selected pole while keeping the contour fixed. For the holomorphic matrix family

$$A(\varepsilon) = \begin{pmatrix} 1+\varepsilon & 1 \\ 0 & 3 \end{pmatrix}, \qquad |\varepsilon| < 1/4,$$

the same residue calculation gives

$$P(\varepsilon) = \begin{pmatrix} 1 & -(2-\varepsilon)^{-1} \\ 0 & 0 \end{pmatrix}, \qquad A(\varepsilon)P(\varepsilon) = (1+\varepsilon)P(\varepsilon).$$

The moving pole stays inside the circle, and the pole at three stays outside. The projection has rank one throughout this parameter disk. The selected eigenvalue is holomorphic, as is the projection, although an arbitrary choice of a normalised eigenvector need not be holomorphic. The theorem below constructs a holomorphic vector without imposing a unit-norm condition.

Theorem 18.2. *Let C be a positively oriented circle disjoint from $\sigma(A)$. Its* Riesz projection *is*

$$P = -\frac{1}{2\pi\mathrm{i}} \int_C R_A(z)\,\mathrm{d}z.$$

It satisfies $P^2 = P$ and $AP = PA$.

Proof. First deform the contour to two nested circles. Then the resolvent identity reduces the product of two integrals to scalar Cauchy integrals. Compactness of C and the inverse series give a thin spectrum-free annulus about it. Choose nested circles C_- and C_+ there. Deformation gives the same P on each. Use the resolvent identity in P^2, with z on C_+ and v on C_-:

$$P^2 = \frac{1}{(2\pi\mathrm{i})^2} \int_{C_+} \int_{C_-} \frac{R_A(z) - R_A(v)}{z - v}\,\mathrm{d}v\,\mathrm{d}z.$$

The term containing $R_A(z)$ integrates to zero in v, because z is outside C_-. Reverse the order for the other term. The integral of $1/(z-v)$ on C_+ is $2\pi\mathrm{i}$. What remains is $-(2\pi\mathrm{i})^{-1}\int_{C_-} R_A(v)\,\mathrm{d}v = P$. Interchanging integrals follows from uniform approximation by finite sums. Finally $AR_A(z) = R_A(z)A = I + zR_A(z)$, so A commutes with P. ∎

For a scalar operator $A = \lambda I$, the resolvent is $-(z-\lambda)^{-1}I$. If the circle encloses λ, its positive contour integral is $-2\pi\mathrm{i}I$, so the minus sign gives $P = I$. If the circle does not enclose λ, the integral and projection are zero.

Theorem 18.3. *Let $A(\varepsilon)$ be a norm-holomorphic bounded-operator family near $\varepsilon = 0$. Suppose a circle C, disjoint from $\sigma(A(0))$, encloses an isolated spectral value λ_0 of $A(0)$ and no other spectrum, and its Riesz projection has one-dimensional range. Then near zero there is a holomorphic eigenvalue $\lambda(\varepsilon)$ with $\lambda(0) = \lambda_0$, and a nonzero holomorphic eigenvector. It is the only eigenvalue inside that circle.*

Proof. Put $A_0 = A(0)$ and $R_0(z) = R_{A_0}(z)$, and write the assumed series as

$$\begin{aligned} E(\varepsilon) &= A(\varepsilon) - A_0 \\ &= \sum_{j\ge 1} \varepsilon^j A_j. \end{aligned}$$

On a smaller disk this series is absolutely norm convergent. Indeed, convergence at a larger positive radius r_0 bounds $\|A_j\| r_0^j$; a geometric comparison then proves summability at every $r < r_0$. Let $M = \sup_{z\in C} \|R_0(z)\|$, finite by norm continuity on the circle. Choose $r > 0$ small enough that

$$q = M \sum_{j\ge 1} \|A_j\| r^j < 1.$$

For $|\varepsilon| \le r$ and $z \in C$, factor $A(\varepsilon) - zI = (I + E(\varepsilon)R_0(z))(A_0 - zI)$. The inverse is

$$\begin{aligned} &R_{A(\varepsilon)}(z) \\ &\quad = \sum_{n\ge 0} R_0(z)\,[-E(\varepsilon)R_0(z)]^n. \end{aligned} \tag{18.1}$$

Thus the circle remains disjoint from the perturbed spectrum, and Theorem 18.2 applies to its projection. Expand each product using the absolutely convergent series for E. The sum of all term norms, with $|\varepsilon|$ replaced by r, is bounded uniformly on C by $\sum_{n\ge 0} Mq^n = M/(1-q)$. Thus the terms can be grouped by powers of ε:

$$\begin{aligned} R_{A(\varepsilon)}(z) &= \sum_{m\ge 0} \varepsilon^m B_m(z), \\ \sum_{m\ge 0} r^m \sup_{z\in C} \|B_m(z)\| &\le \frac{M}{1-q}. \end{aligned}$$

Each coefficient is a finite sum of products of $R_0(z)$ and the A_j, so it is continuous on C. Uniform absolute convergence permits termwise integration of this series. In particular,

$$\begin{aligned} P(\varepsilon) &= \sum_{m\ge 0} \varepsilon^m P_m, \\ P_m &= -\frac{1}{2\pi\mathrm{i}} \int_C B_m(z)\,\mathrm{d}z. \end{aligned}$$

The integral norm bound proves absolute norm convergence of this last series. To re-expand at an interior centre ε_*, choose $\delta > 0$ with $|\varepsilon_*| + \delta < r$. The binomial expansion has absolute bound

$$\sum_{m\geq 0} \|P_m\| \, (|\varepsilon_*| + \delta)^m < \infty.$$

It can therefore be regrouped into a local norm power series. This proves norm holomorphy throughout the parameter disk. By shrinking that disk we have $\|P(\varepsilon) - P(0)\| < 1$. If two projections P, Q differ by norm less than one, the map $P : \operatorname{ran} Q \to \operatorname{ran} P$ is injective: $Px = 0$ and $Qx = x$ imply $\|x\| \leq \|Q - P\| \, \|x\|$, hence $x = 0$. Interchanging P and Q proves that their ranges have the same dimension when one is finite-dimensional. Thus the new range is also a line.

Choose nonzero u_0 in the old range and a continuous linear functional ℓ with $\ell(u_0) = 1$; for example use an inner product with u_0, normalised. The series for P applied to u_0 is an absolutely norm-convergent vector power series. Thus $u(\varepsilon) = P(\varepsilon)u_0$ is holomorphic and nonzero near zero. Commutation makes its line invariant under $A(\varepsilon)$. The scalar on that line is

$$\lambda(\varepsilon) = \frac{\ell(A(\varepsilon)u(\varepsilon))}{\ell(u(\varepsilon))}.$$

The Cauchy product for $A(\varepsilon)u(\varepsilon)$ converges absolutely in norm, since its sum of term norms is bounded by the product of the two separate sums. Applying the continuous functional ℓ gives scalar holomorphic power series in numerator and denominator. The denominator equals one at zero. Its reciprocal is holomorphic by the scalar geometric series, so this quotient is holomorphic. Since $Pu = u$, applying the contour formula to this eigenvector shows that $\lambda(\varepsilon)$ lies inside the circle: the scalar integral is one inside and zero outside. At zero it must be λ_0. Every other eigenvector with eigenvalue inside the circle also satisfies $Px = x$, and therefore lies on the same line. Its eigenvalue must coincide with $\lambda(\varepsilon)$. ∎

The rank-one projection hypothesis expresses algebraic simplicity. A one-dimensional eigenspace alone does not suffice, even for a matrix with a nontrivial Jordan block.

A one-dimensional eigenspace need not give a rank-one contour projection. Consider the Jordan matrix

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad R_J(z) = \begin{pmatrix} (1-z)^{-1} & -(1-z)^{-2} \\ 0 & (1-z)^{-1} \end{pmatrix}.$$

Its eigenspace at one consists of vectors with second coordinate zero. It is a line. Nevertheless, a circle around one gives the projection I: the double-pole entry has zero residue, and both diagonal entries contribute one after the contour sign. The projection therefore has rank two. The simple-branch hypothesis rules out this possibility.

18.3 Herglotz functions and a supplied density

A *Herglotz function* is holomorphic on $\mathbb{H}$ with nonnegative imaginary part.

Proposition 18.4. *For bounded self-adjoint A and $h \in \mathcal{H}$, the scalar function $z \mapsto \langle h, R_A(z)h\rangle$ is a Herglotz function.*

Proof. Holomorphy follows from the norm-holomorphic resolvent. Put $u = R_A(z)h$. Self-adjointness gives

$$\operatorname{Im} \langle h, R_A(z)h\rangle = \operatorname{Im} \langle (A - zI)u, u\rangle = \operatorname{Im} z \, \|u\|^2 \geq 0. \tag{18.2}$$

The positive sign uses antilinearity in the first slot. ∎

Theorem 18.5 (Stieltjes inversion for a supplied density). *Let $\rho \in L^1(\mathbb{R})$ be nonnegative and define its Cauchy transform by*

$$F(z) = \int_{\mathbb{R}} \frac{\rho(t)}{t - z}\,\mathrm{d}t, \qquad z \in \mathbb{H}.$$

This is a Herglotz function. For every real $c < d$,

$$\lim_{y \downarrow 0} \frac{1}{\pi} \int_c^d \operatorname{Im} F(x + \mathrm{i}y)\,\mathrm{d}x = \int_c^d \rho(t)\,\mathrm{d}t. \tag{18.3}$$

Proof. The parameter-integral theorem, Theorem 10.9, proves holomorphy. The imaginary kernel is $y/((t-x)^2+y^2)$ for $z = x+\mathrm{i}y$, which is positive. Tonelli (M4 in § 4.1) and integration in x give

$$\begin{aligned}\frac{1}{\pi} \int_c^d \operatorname{Im} F(x + \mathrm{i}y)\,\mathrm{d}x &= \int_{\mathbb{R}} \frac{\rho(t)}{\pi} \Big[\arctan \frac{d - t}{y} \\ &\qquad - \arctan \frac{c - t}{y}\Big]\,\mathrm{d}t.\end{aligned}$$

The bracket divided by π lies between zero and one. As $y \downarrow 0$ it tends to the indicator of (c, d) except for the two endpoints. These have Lebesgue measure zero. Dominated convergence, Theorem 3.5, proves the stated recovery formula. ∎

This recovers the supplied density's interval integrals. No general representing-measure or operator spectral-measure theorem is being asserted.

18.4 A bounded Friedrichs model

The model couples one discrete level to a continuous range of energies. The scalar component is its level amplitude. The function component describes the continuum. With zero coupling these evolve independently; the off-diagonal terms allow exchange between them. The coupling function specifies how strongly each continuum energy participates. Its squared modulus is the density entering the scalar resolvent. We ask where the continued scalar pole lies after this coupling is switched on. As in Lecture 12, we parametrise the pole by a decay rate Γ. Its energy width is $\hbar\Gamma$, with fixed $\hbar > 0$. This parametrisation does not assert an exact decay law.

Fix $0 < a < \lambda_0 < b$ and a real coupling g. On $\mathcal{H} = \mathbb{C} \oplus L^2([a, b])$, take $v \in L^2([a, b])$ and define

$$H_g(c, f) = \left(\lambda_0 c + g\,\langle v, f\rangle\,,\; g c\, v + \lambda f\right).$$

Here $(\lambda f)(\lambda) = \lambda f(\lambda)$, and the direct-sum norm is $(|c|^2 + \|f\|_2^2)^{1/2}$. Multiplication by λ is bounded by b. Cauchy–Schwarz bounds the off-diagonal map by $|g|\,\|v\|_2$. The diagonal map is bounded by $\max(\lambda_0, b)$. Thus H_g is bounded. Expanding its inner product shows self-adjointness: the cross terms are $g\bar{c}\,\langle v, h\rangle$ and $g d\,\langle f, v\rangle$ for arguments (c, f) and (d, h), and they interchange under conjugation. The real multiplication terms have the same property. The density is $\rho(\lambda) = |v(\lambda)|^2 \in L^1([a, b])$.

Theorem 18.6. *For $z \notin \mathbb{R}$ and the level vector $e_0 = (1, 0)$,*

$$m_g(z) := \langle e_0, R_{H_g}(z) e_0 \rangle = \frac{1}{\lambda_0 - z - g^2 F(z)}, \qquad F(z) = \int_a^b \frac{\rho(\lambda)}{\lambda - z}\, \mathrm{d}\lambda. \tag{18.4}$$

Proof. Solve $(H_g - zI)(c, f) = (d, h)$. The second component, whose multiplication inverse has norm at most $1/|\operatorname{Im} z|$, gives $f = (h - gcv)/(\lambda - z)$. Substitute into the first component:

$$D_g(z) c = d - g \int_a^b \frac{\overline{v} h}{\lambda - z}\, \mathrm{d}\lambda, \qquad D_g(z) = \lambda_0 - z - g^2 F(z).$$

For $\operatorname{Im} z > 0$, the imaginary part of $D_g(z)$ is $-\operatorname{Im} z - g^2 \operatorname{Im} F(z) < 0$. The lower half-plane has the opposite sign. Hence the denominator is nonzero. The displayed equations construct a unique solution for every (d, h) and bound it by a constant times $|d| + \|h\|_2$. They therefore give the bounded two-sided inverse. For $(d, h) = e_0$, its first component is $1/D_g(z)$, proving the formula. ∎

For example, take $[a, b] = [1, 3]$, $\lambda_0 = 2$ and $v(\lambda) = \sqrt{\lambda}$. At $g = 1/2$ and $z = 0$ the same inverse calculation is valid: $F(0) = \int_1^3 1\, \mathrm{d}\lambda = 2$ and $D_g(0) = 2 - 1/2 = 3/2$. Thus $m_g(0) = 2/3$. The bath inverse is bounded there, and the nonzero Schur denominator verifies invertibility at this real point too.

18.5 The continued pole and its decay rate

Assume that ρ agrees almost everywhere on a closed interval $I \subset (a, b)$ about λ_0 with the restriction of a function holomorphic on a complex neighbourhood of I, also denoted by ρ. Assume $\rho(\lambda_0) > 0$ for this representative. Shrink I about λ_0 so that its closed lower half-disk lies in that neighbourhood, as required by Theorem 12.2. Extend ρ by zero outside $[a, b]$ for the real-line integral. The proved deformation theorem, Theorem 12.2, continues F from above through that interval to a holomorphic F_{II}. Its jump and reality give $\operatorname{Im} F_{\mathrm{II}}(\lambda_0) = \pi \rho(\lambda_0)$. Continue the scalar m_g as $(\lambda_0 - z - g^2 F_{\mathrm{II}}(z))^{-1}$. This continuation concerns a scalar function, not an operator-norm inverse through the bath spectrum.

Theorem 18.7. *For sufficiently small real nonzero g, the continued scalar resolvent has a unique pole in a fixed disk about λ_0. It is simple and has location*

$$z_g = \lambda_0 - g^2 F_{\mathrm{II}}(\lambda_0) + O(g^4) = E_{\mathrm{res}} - \mathrm{i}\hbar\Gamma/2, \qquad \Gamma = \frac{2\pi g^2 \rho(\lambda_0)}{\hbar} + O(g^4) > 0,$$

where $\hbar > 0$ is fixed and $E_{\mathrm{res}} = \operatorname{Re} z_g$.

Proof. Choose a closed disk of radius r within the continuation domain and bounds $|F_{\mathrm{II}}| \le M$ and $|F_{\mathrm{II}}'| \le B$ there. For $|g|^2 M < r$ and $|g|^2 B < 1$, the map $z \mapsto \lambda_0 - g^2 F_{\mathrm{II}}(z)$ is a contraction of that disk. The contraction theorem gives a unique fixed point z_g. Its iterates are holomorphic in complex g and converge locally uniformly by a geometric bound. Thus z_g is holomorphic in g. The fixed-point equation first gives $z_g - \lambda_0 = O(g^2)$. The derivative bound then gives the displayed expansion with error $O(g^4)$. At the zero, the denominator derivative is $-1 - g^2 F_{\mathrm{II}}'(z_g) \neq 0$. The reciprocal therefore has a simple pole, with residue the reciprocal of this derivative. Taking imaginary parts of the expansion proves the decay-rate formula and its positivity for small nonzero real g. ∎

The diagram shows the scalar continuation and its lower pole.

Self-adjointness excludes physical nonreal eigenvalues. The continued pole does not assert an exact exponential survival law.

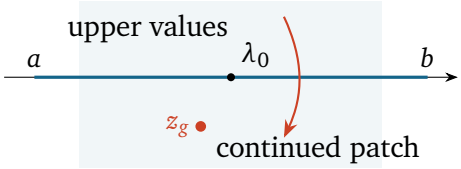


Figure 18.2: Local continuation of the scalar resolvent through an interior bath interval. The lower continued pole is not a nonreal eigenvalue of H_g.

18.6 The compact spectral input

An operator is *compact* if every bounded sequence has a subsequence whose images converge in norm.

Theorem 18.8 (Compact self-adjoint spectral theorem · *cited*)**.** *A compact self-adjoint operator has an orthonormal basis of eigenvectors, including a basis of its kernel. Its nonzero eigenvalues are real, have finite multiplicity, and can accumulate only at zero.*

This is an external input, from Axler, Theorem 10.106 with 10.82 and 10.93. None of the resolvent or bounded-model proofs above used it. The general completeness deduction in Lecture 24 uses it and inherits that qualification.

Summary.

- Bounded resolvents are norm holomorphic. With $(A - zI)^{-1}$ the Riesz projection carries a minus sign. For a norm-holomorphic bounded family, an isolated rank-one contour projection gives a holomorphic simple eigenvalue branch.
- A supplied nonnegative integrable density is recovered by Stieltjes inversion of its Cauchy transform.
- The bounded Friedrichs Schur formula gives a scalar continuation. For a locally analytic positive density its simple lower pole has the proved leading decay rate $2\pi g^2 \rho(\lambda_0)/\hbar$.
- Compact self-adjoint spectral completeness remains a cited input.

References

G. Teschl, *Mathematical Methods in Quantum Mechanics*, 2009, pp. 73–79 and Theorem 3.21, p. 108; only bounded resolvents and supplied-density inversion are used here. S. Axler, *Measure, Integration & Real Analysis*, 2025 version, Theorems 10.82, 10.93 and 10.106, pp. 318, 322, 329. The Friedrichs Schur derivation and local pole proof are given here in full.